

Homework — 2.1.2 Thinking Ahead

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 2.1.2 lesson.

Part A — This week's topic (~14 marks)

1. A library system has one routine that **issues** a book to a member and another that marks a book as **returned**. Complete the table: classify each item as an **Input**, an **Output** or a **Precondition** of the routine named. [3] (AO2)

Item	Input / Output / Precondition
The member's ID scanned at the desk (issue routine)	
The updated loan record written to the database (issue routine)	
"The member has no unpaid fines" (issue routine)	
The book's barcode scanned at the desk (issue routine)	
The printed receipt showing the due date (issue routine)	
"The book is currently on loan to this member" (return routine)	

2. A train-times website **caches** the popular "London → Manchester" timetable. The jumbled steps below describe how the cache serves two users. Number them **1–6** in the correct order. [2] (AO2)

Step	Order (1–6)
The computed timetable is stored in the cache.	
A user requests the London → Manchester timetable.	
The system checks the cache — the timetable is not there (a cache miss).	
A second user requests the same timetable and is served the stored copy straight from the cache.	
The timetable is computed by querying the live database.	
The timetable is returned to the first user.	

3. A developer proposes this caching design for the same website: *"Compute the full London → Manchester timetable once at midnight, store it in the cache, and serve that copy to every user all day. Never refresh it before the next midnight."*

(a) State **one** benefit this design still provides. [1] (AO2)

(b) Explain the **flaw** in the design, referring to what happens if a train is delayed during the day. [2] (AO2)

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(c) Suggest **one** improvement to the design that keeps most of the benefit while reducing the flaw. [1] (AO2)

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4. A games studio is building three different games that each need a feature to load and save the player's progress to a file. Give **two** benefits of writing this as a single **reusable component** rather than coding it separately in each game. [2] (AO2)

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5. A developer is planning a routine that processes an online food order. Using *thinking ahead*, state for this routine: **one** input, a **precondition** that must hold before payment is taken, and **one** thing it would be useful to **cache**. [3] (AO2)

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Part B — Retrieval: keep it fresh (~6 marks)

6. (2.1.1) State **one** detail a developer would **keep** and **one** they would **discard** when abstracting a "student" for a school timetabling system. [2] (AO2)

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7. (1.2.1) State what an **interrupt** is and one reason an operating system uses interrupts. [2] (AO1)

8. (1.1.1) State **two** factors that affect the **performance** of a processor. [2] (AO1)
